

```
In [1]: import numpy as np
```

```
In [2]: a=np.array([[10,20],[30,40]])  
b=np.array([[1,2],[3,4]])
```

```
In [3]: print(a)
```

```
[[10 20]  
 [30 40]]
```

```
In [4]: print(b)
```

```
[[1 2]  
 [3 4]]
```

```
In [6]: #matrix addition  
c=np.add(a,b)  
d=a+b  
print(c)  
print("-----OR-----")  
print(d)
```

```
[[11 22]  
 [33 44]]  
-----OR-----  
[[11 22]  
 [33 44]]
```

```
In [9]: #matrix subtraction
print(a)
print("-----")
print(b)
c=np.subtract(a,b)
d=a-b
print("-----")
print("Result")
print("-----")
print(c)
print("-----OR-----")
print(d)
```

```
[[10 20]
 [30 40]]
-----
[[1 2]
 [3 4]]
-----
Result
-----
[[ 9 18]
 [27 36]]
-----OR-----
[[ 9 18]
 [27 36]]
```

```
In [10]: #matrix Multiplication---Elementwise Matrix Multiplication
print(a)
print("-----")
print(b)
c=np.multiply(a,b)
d=a*b
print("-----")
print("Result")
print("-----")
print(c)
print("-----OR-----")
print(d)
```

```
[[10 20]
 [30 40]]
-----
[[1 2]
 [3 4]]
-----
Result
-----
[[ 10  40]
 [ 90 160]]
-----OR-----
[[ 10  40]
 [ 90 160]]
```

In [11]: *#matrix Multiplication---actual Matrix Multiplication*

```
print(a)
print("-----")
print(b)
c=np.matmul(a,b)
d=np.dot(a,b)
print("-----")
print("Result")
print("-----")
print(c)
print("-----OR-----")
print(d)
```

```
[[10 20]
 [30 40]]
```

```
[[1 2]
 [3 4]]
```

Result

```
[[ 70 100]
 [150 220]]
```

-----OR-----

```
[[ 70 100]
 [150 220]]
```

In [12]: *#matrix division*

```
print(a)
print("-----")
print(b)
c=np.divide(a,b)
d=a/b
print("-----")
print("Result")
print("-----")
print(c)
print("-----OR-----")
print(d)
```

```
[[10 20]
 [30 40]]
```

```
[[1 2]
 [3 4]]
```

Result

```
[[10. 10.]
 [10. 10.]]
```

-----OR-----

```
[[10. 10.]
 [10. 10.]]
```

```
In [13]: #matrix division
print(a)
print("-----")
print(b)
c=np.floor_divide(a,b)
d=a//b
print("-----")
print("Result")
print("-----")
print(c)
print("-----OR-----")
print(d)
```

```
[[10 20]
 [30 40]]
-----
[[1 2]
 [3 4]]
-----
Result
-----
[[10 10]
 [10 10]]
-----OR-----
[[10 10]
 [10 10]]
```

```
In [14]: #matrix Modulo division
print(a)
print("-----")
print(b)
c=np.mod(a,b)
d=a%b
print("-----")
print("Result")
print("-----")
print(c)
print("-----OR-----")
print(d)
```

```
[[10 20]
 [30 40]]
-----
[[1 2]
 [3 4]]
-----
Result
-----
[[0 0]
 [0 0]]
-----OR-----
[[0 0]
 [0 0]]
```

```
In [15]: #matrix Exponentiation
print(a)
print("-----")
print(b)
c=np.power(a,b)
d=a**b
print("-----")
print("Result")
print("-----")
print(c)
print("-----OR-----")
print(d)
```

```
[[10 20]
 [30 40]]
-----
[[1 2]
 [3 4]]
-----
Result
-----
[[      10      400]
 [ 27000 2560000]]
-----OR-----
[[      10      400]
 [ 27000 2560000]]
```

In []: